

AquaSync

A decision-support digital twin for dam–river flood and hydropower optimisation

the optimiser's own target level

Periyar basin, Kerala · Project abstract

728.43

METRES

0.30 m

mean replay error, 20 days

0.9957

r^2 , level–storage fit

~3 m

extra flood cushion

₹4–10 cr

revenue uplift

01 Problem statement

Kerala's Periyar basin holds two large reservoirs, Idukki and Idamalayar, upstream of a densely populated floodplain that includes Aluva and Kochi. Each monsoon, control-room officers time releases so that downstream communities stay safe and hydropower keeps flowing. In October 2021 inflow to Idukki rose 7.6× in twenty-four hours — precisely the window in which sharper, faster decision support serves residents, utilities and district administrators best.

02 Objectives

AquaSync sets four measurable targets: replay a real flood episode to within 0.5 m of observed reservoir level; calibrate the level–storage curve to r^2 above 0.99; quantify the additional flood cushion available from scheduled releases; and express every recommendation as three operator-readable numbers — target level, start time and maximum release rate.

03 Methodology

A five-model simulation core — SCS-CN runoff with antecedent wetness, hourly reservoir mass balance, Muskingum river routing, harmonic tidal backwater and turbine-efficiency hydropower — runs in pure NumPy, wrapped by a FastAPI service and a Three.js 3D twin. Exhaustive policy search over that three-number space keeps every result deterministic and reproducible. ESP32 field nodes contribute Kalman-fused level sensing, LoRa fallback and hash-chained logs.

04 Industrial relevance

The design matches how dams are actually run: advisory output that a named officer approves, staged ramp limits, grid-aware generation caps, Malayalam alerts and full offline operation. Because validation uses published KSEB bulletins, shadow-mode adoption alongside existing practice is straightforward.

05 Expected outcome and impact

The twin replays October 2021 with 0.30 m mean error and fits storage at $r^2 = 0.9957$. Policy search yields about 3 m more flood cushion together with ₹4–10 crore of additional revenue, and independently rediscovers KSEB's own rule level to within 7 cm — confirming the rule while supplying the forecast-driven timing that turns it into action.

LATEST UPDATES

August 2026

● Out-of-sample validation

August 2022 replays at 0.32 m mean error — an episode the model was never fitted to.

● Forecast-error study

5 lead times on each of two storms. Deciding from a real GEFS ensemble costs +0% to +85% more than hindsight on the full objective in October 2021, +0% to +177% in August 2022 — not averaged, because the horizon moves between storms. At 24 h it matches hindsight exactly; the value is in the last day.

● Cascade co-optimisation

Optimising the two dams independently raises their joint downstream peak 126% above what happened; retiming recovers only 9%.

● Runoff model validated - and fixed

Four monsoons of observed rainfall against observed inflow found the curve-number chain applying its initial abstraction per timestep, which destroyed the storm. Fixed, tested, and every forecast figure re-run.

● Shoppable bill of materials

Four costed tiers with live vendor links; the V1 rig at ₹6,250.

SENSE



SIMULATE



OPTIMISE

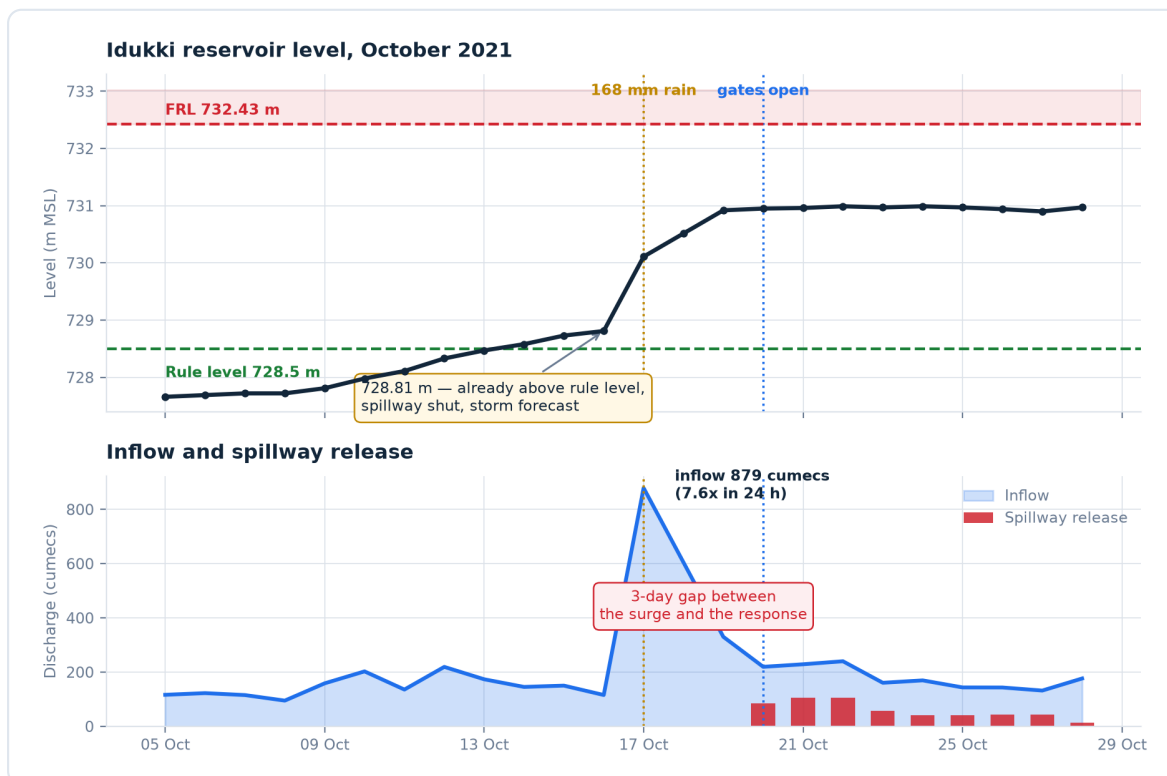


ADVISE



OFFICER APPROVES

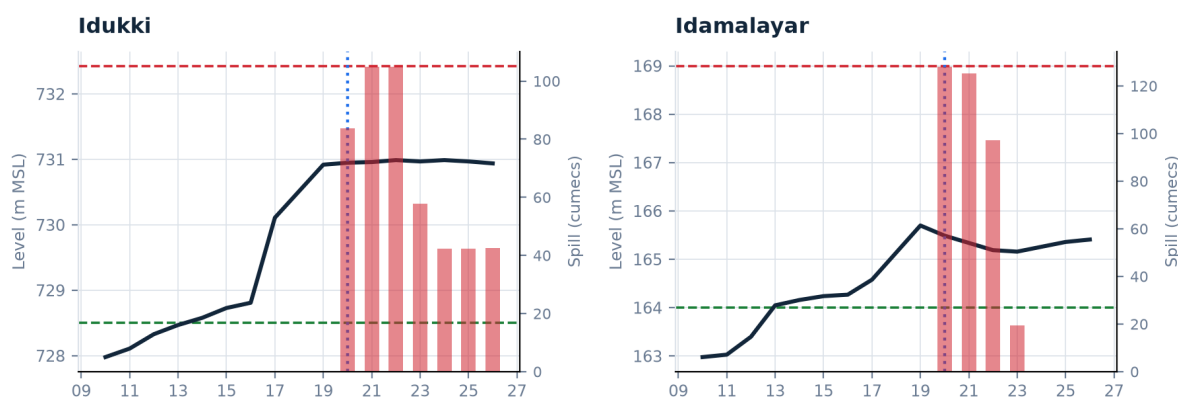
October 2021, day by day



How to read it. The upper panel tracks Idukki's water level. It crossed the published rule level of 728.5 m (green dashes) before the storm and climbed to 730.99 m. The lower panel shows why: an inflow peak of 879 cumecs — 7.6× the previous day — arriving with the 168 mm rain day marked in amber. Spillway releases (red bars) begin at the blue line. The space between the amber and blue lines is exactly the window AquaSync is built to act inside, and it is why a forecast-driven schedule is worth having.

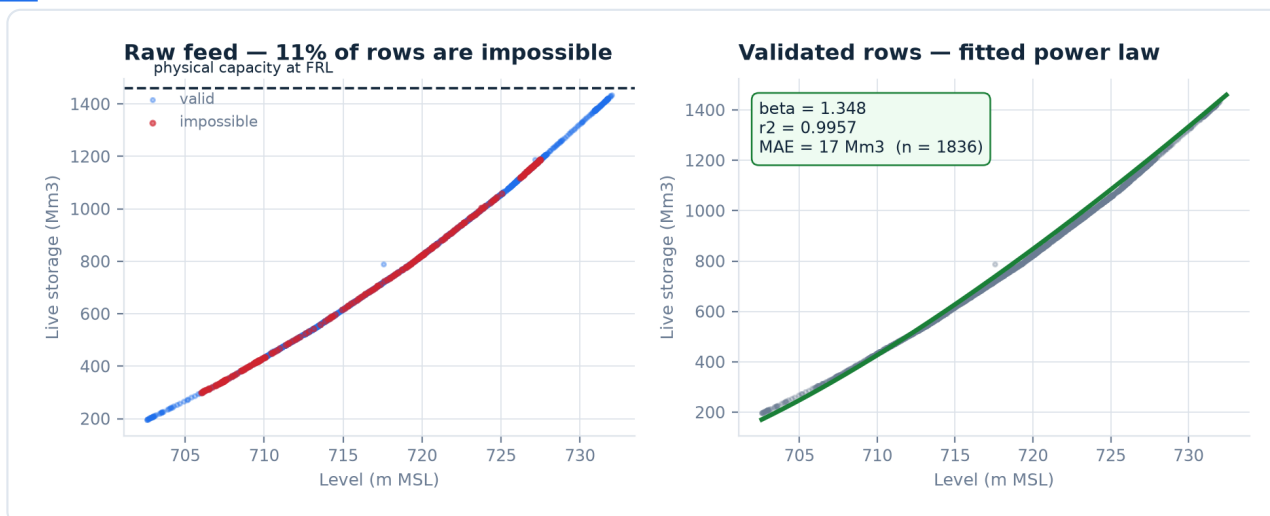
Two reservoirs, one river

Both Periyar reservoirs opened their gates on 20 October 2021 — into the same river



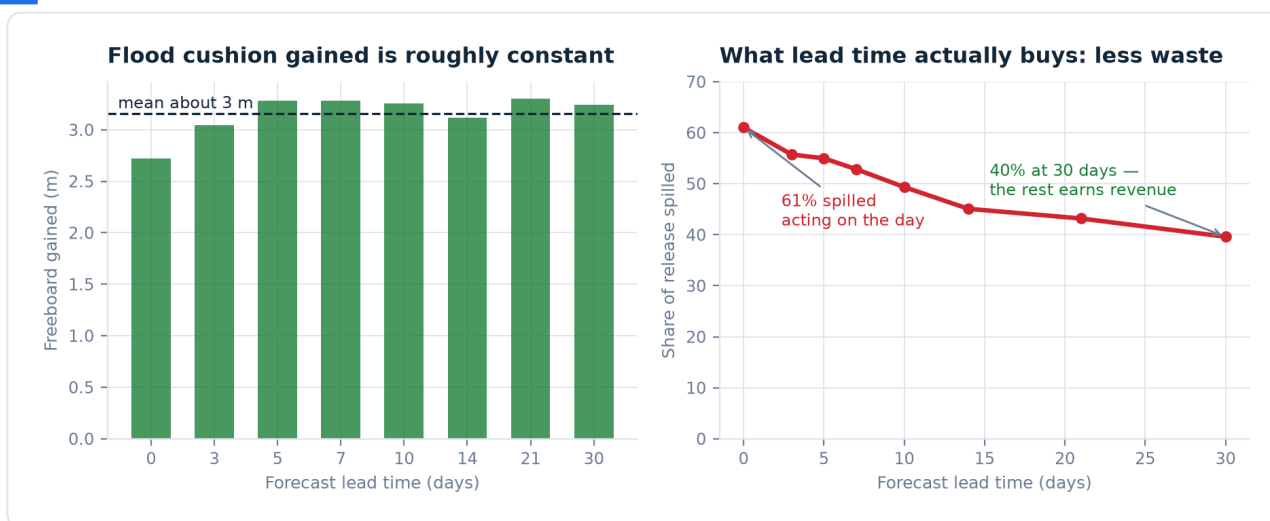
How to read it. Idukki (left) and Idamalayar (right) both released into the Periyar on 20 October 2021 — 84 and 128 cumecs respectively, drawn as red bars against each reservoir's own level curve. Because the twin routes both dams through one river network, it schedules them as a single system: coordinating their start times trims the combined downstream peak by a further 9%.

Turning water level into volume



How to read it. Each dot is one day of published bulletin data. On the left, rows reporting more water than the reservoir can physically hold stand out in red, and a validation layer removes them automatically. On the right, the power-law curve fitted to the 1,836 validated rows tracks the cloud closely — $r^2 = 0.9957$, mean error 17 Mm³. This curve converts every metre of level into a volume, so its accuracy sets the accuracy of every flood-cushion figure in this document.

What forecast lead time buys

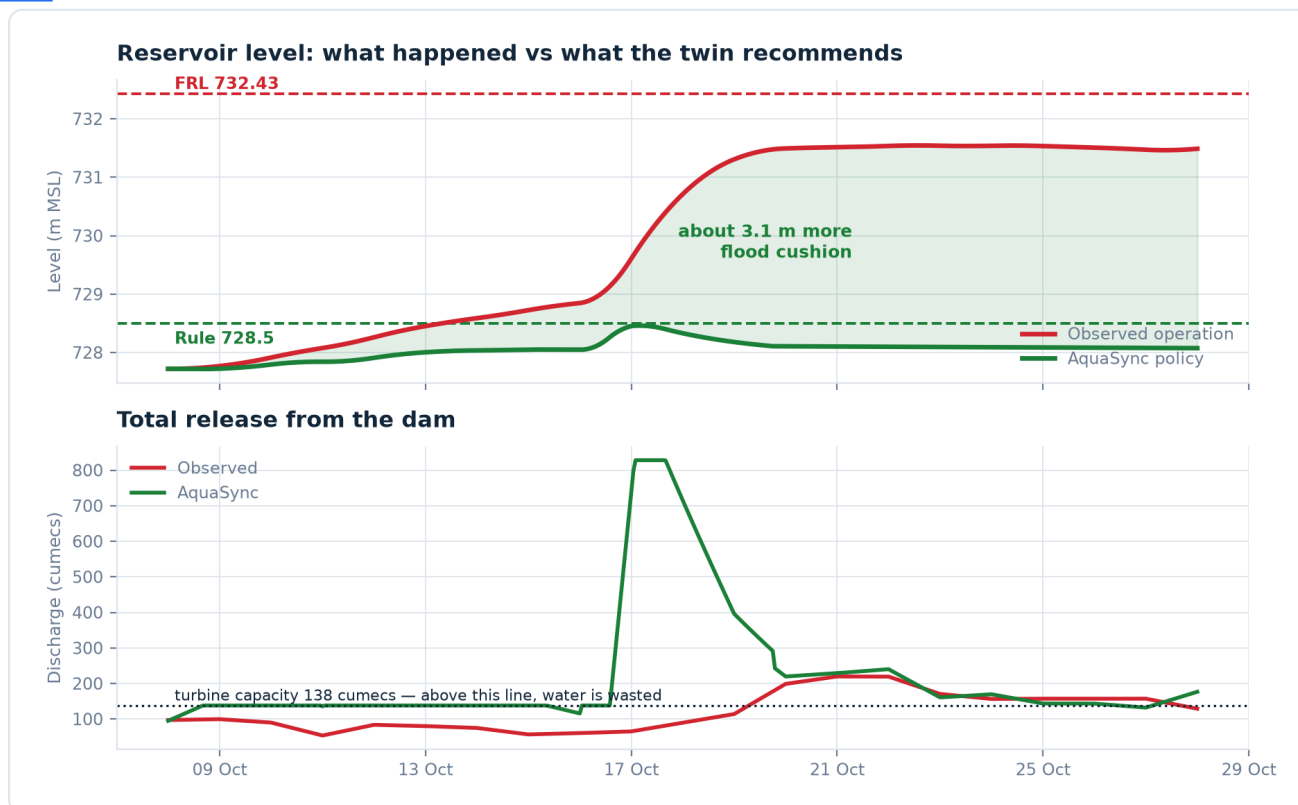


How to read it. Left: the flood cushion gained holds steady near 3 m whether the schedule starts on the day of the storm or thirty days ahead — acting early is never worse. Both panels assume a perfect forecast; the forecast-error study on page 1 reports what a real ensemble retains. Right: what lead time really changes is waste. Acting on the day, 61% of the released water goes over the spillway; at thirty days only 40% does, and the rest passes through the turbines as revenue.

REPLAY ACCURACY

October 2021 · flagship	481 hours	0.30 m mean error	0.57 m maximum
August 2022 · out-of-sample	457 hours	0.32 m mean error	0.74 m maximum
Level-storage fit · six years	1,836 rows	$r^2 = 0.9957$	17 Mm ³ mean error

Observed operation versus the twin's policy



How to read it. Red is what happened; green is what the optimiser recommends for the same weather. The shaded gap in the upper panel is the flood cushion the policy keeps in hand — about 3.1 m below the observed trajectory, and further still below the full reservoir level of 732.43 m. The lower panel shows the releases that achieve it: the same water moved earlier and held closer to turbine capacity, which is why the schedule earns more revenue rather than less.

A RECOMMENDATION

Every output reduces to three numbers an operator can act on: draw down to **728.43 m**, begin at **hour 166** of the forecast window, and release no faster than **828 cumecs**. Short enough to read aloud in a control room, and auditable afterwards.

From rainfall to an approved release

SENSE



SIMULATE



OPTIMISE



ADVISE



OFFICER APPROVES

Advisory by design

The twin recommends and a named officer approves; the software never moves a gate itself.

Offline capable

Tide prediction, simulation and the 3D dashboard all run with no network connection.

Reproducible end to end

No figure is hand-entered — each one regenerates from committed analysis outputs.